

ReachNT

Remote repair triage that ranks by need, shares trips with H3 hexagons, and makes someone sign for who waits

Data analysis report

CDU IT Code Fair 2026, Artificial Intelligence Challenge: Housing maintenance triage

Team AIC015 · Top Enders

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We acknowledge the Larrakia people, Traditional Owners of the land on which Charles Darwin University's Darwin campuses stand, and the Traditional Owners of all the Country this report discusses. This report contains no images, names or voices of people. All repair requests in it are synthetic.

Summary

Remote NT public housing has too few trades and very long distances; an emergency job in a very remote community can cost several times the same job in town, mostly in travel [6]. A schedule built to clear the most jobs per dollar keeps choosing town, and nobody decides that remote tenants wait.

ReachNT ranks fault reports by need alone: never by distance, cost, when or how a tenant reported, how much they said, or their English. Whoever the tenant tells asks the same short questions. ReachNT plans weekly trips around crew hours, road closures and airstrips, using Uber's H3 hexagonal grid to let one tradesperson serve neighbouring communities on one trip. It shows what each setting costs and who waits, and a named person signs one. A person can change urgency any time; a missed visit is reassigned without losing its place.

On real NT geography with synthetic requests (70 communities, 8,345 houses, 37,508 requests), planning for the cheapest jobs cost \$559 per repair, but 9 in 10 urgent remote repairs took up to 62 days, against 3 in town. ReachNT cost \$803 (44% more), fixed 9 in 10 urgent remote repairs within 3 days, and cut the days households lived with faults by 73%. Shared trips saved \$73 per repair. The standard questions cut the points lost for saying little from 19 to 2. We recommend DHLGCD publish its triage rules, ask every tenant the same questions, record who accepts each trade-off, and co-design with communities first.

1 Introduction

1.1 The problem

The NT Government manages 5,058 remote public housing dwellings in 73 communities [1]. In December 2022, 55% of remote houses were overcrowded [2]. In 2022–23, 54% of First Nations households in very remote areas had major structural problems, against 25% in major cities [4]. Tenants report faults by phone (1800 104 076) or to a local Community Housing Officer, and the Department sorts each job into one of three categories with a response time [5]:

Table 1. NT response times for public housing repairs (DHLGCD Fact sheet FS17, version 10/25).

Category	Town	Remote
Immediate (dangerous)	4 hours	4 hours; a Remote Housing Maintenance Officer makes it safe if no contractor is available
Urgent	2 business days	5 business days
Routine	10 business days	25 business days

The published standard already gives remote houses two and a half times as long. Costs explain why: emergency repairs cost 8.5 times more in very remote communities than in remote ones, and travel can be 96% of an emergency job's cost, against 11–37% for planned work [6]. On the APY Lands, Housing SA batched about ten work orders per travel order [7].

In 2015, tenants at Santa Teresa listed more than 600 repairs. The Tribunal (2018) and the Court of Appeal (2022) held that public housing must be at least safe, and in 2023 the High Court allowed compensation for the distress of living in unrepaired homes [11]. The ANAO found that community housing organisations could not tell tenants where their repairs were up to, because they had no access to the IT systems [10].

1.2 Purpose and scope

The brief asks for a tool that reads free-text fault reports, ranks jobs by urgency, safety and logistics, explains each job's place, makes the equity trade-off visible and leaves the decision with a person. We add one rule: urgency and logistics are separate steps. Urgency is about the tenant's need; logistics is about getting a tradesperson there. A single priority score that mixes them is how distance gets into "priority" without anyone choosing it.

The scope is responsive repairs in the 70 remote communities listed in the 2023 review of the NT remote housing partnership [2] and in the five towns their trades work from. Planned maintenance, homelands and town camps are out of scope.

1.3 Who it is for

The ReachNT portal has three views of the same record: the regional coordinator (queue, trips, checks, the trade-off and sign-off), the tradesperson (a run sheet that works without signal), and the tenant (a repair tracker and a text message a Community Housing Officer or interpreter can read out). Reports come in through whoever the tenant tells: repairs-line staff, the Community Housing Officer, the maintenance officer, a tradesperson, or the tenant in the app. The coordinator does not enter them; they check urgency and decide who goes when a visit misses.

2 Methodology

2.1 Data

No public dataset of NT repair requests exists; the Menzies evaluation of Healthy Homes could not obtain job-level data either [9]. We built a real setting and filled it with synthetic requests. The 70 communities and their 4,481 houses come from the 2023 NPRH review [2], coordinates from the NTG communities list, the 3,864 hub-town houses from the 2021 Census, wet-season closures from the NT road-restriction register, airstrips from OurAirports, response clocks from FS17 [5] and s 63 [16], health order from Healthabitat [8], and the job mix and cost structure from Grealy et al. [7] and Nous [6] (Table C1). Every output carries a "synthetic" label.

Each community belongs to the nearest of five crew bases (Darwin, Katherine, Tennant Creek, Alice Springs, Nhulunbuy) by travel time. Access bands follow Nous: more than 2 hours is remote, more than 6 very remote. Ten island communities are fly-in only; 23 are cut by road in some wet-season weeks (Figure 1).

2.2 Reading a fault report

Phrase rules, written from the FS17 examples and the s 63 emergency list, match 23 fault types (including damp and mould, which England's 2025 Awaab's Law singles out) and four modifiers (vulnerable occupant, crowded house, repeat report, negation). A TF-IDF and logistic regression model (scikit-learn) catches wording the rules miss. A person reads the report when no rule matches and the model is under 55% sure, when the two disagree on category, when the model gives 25% or more to a dangerous fault, or when a danger word (smoke, melting, sparks, sewage) appears in a reading that is not Immediate. A negated danger ("no sparks now") stays Immediate and goes to a person: the reader never downgrades danger on its own. Both the Python engine and the portal use the same rules file.

2.3 Need-only urgency

Category sets the clock. Inside a category, points order the line:

points = category (1000/500/100) + harm (0–100) + Healthy Living Practice (Safety 40 down to HLP 9 at 4) + who lives there (vulnerable +25, crowded +10) + repeat report (+10) + waiting (+3 a day past half the clock)

Harm values adapt the HHSRS severity weighting [14]. The waiting term means no job waits forever. The only inputs are the fault, what the report says and days waited; a unit test fails if

distance, cost, region or community appear in the function.

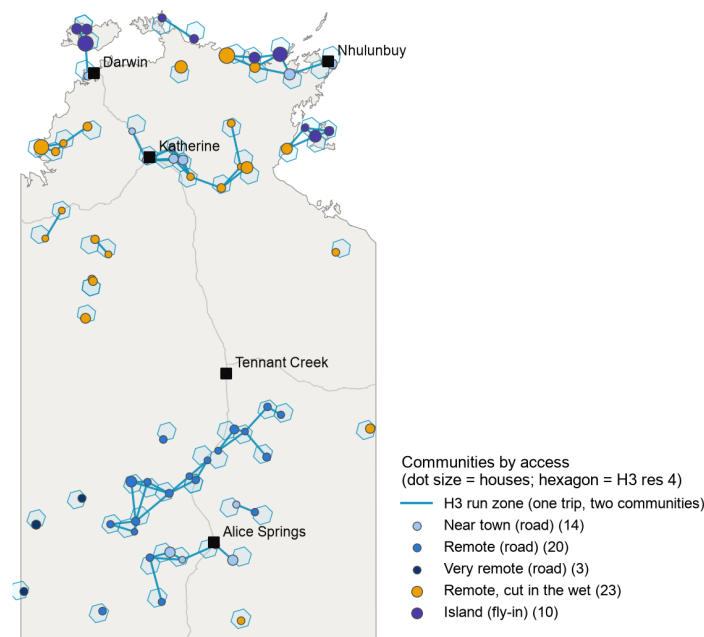


Figure 1. The 70 communities by access band, each community's H3 resolution-4 hexagon, and the 68 possible shared trips (pairs of communities within two hexagon rings that can share a trip).

2.4 H3 hexagons

H3 is Uber's open-source hierarchical grid of hexagons [21]. Every point sits in one cell at each of 16 resolutions, and each finer level is about a seventh of the area. A hexagon has six neighbours, all the same distance from its centre, so "every cell within k rings" is a fair circle in every direction; square grids such as S2 have edge and corner neighbours at two distances [21]. ReachNT uses H3 four ways:

1. **Shared trips.** Two communities served from the same hub share one trip when their resolution-4 cells (about 1,770 km²) are within two rings, about 90 km. That gives 68 pairs, such as Ngukurr and Rittarangu, and the three Tiwi communities.
2. **While you're out there.** A tradesperson's phone lists open jobs in their trade within three resolution-5 rings of the communities on their run.
3. **A house is a hexagon.** Operational records hold a house ID and its resolution-10 cell (a 76 m hexagon), never an address.
4. **Publishing.** Anything shown outside the Department is grouped into resolution-6 cells (36 km²) and hidden below five jobs.

H3 does not know where roads are, so ReachNT pairs it with the road-restriction register and live closures for travel.

2.5 Planning trips

Each week, for each hub and trade, a constraint solver (Google OR-Tools CP-SAT) chooses which communities get a visit, alone or on a shared trip, and which jobs are done. It maximises job value minus $\lambda \times$ cost, where λ is the weight on money, within each tradesperson's 40 hours, travel included. A road trip costs vehicle running plus driving time, a fly-in costs a charter for both legs, and a run zone costs hub $\rightarrow$ A $\rightarrow$ B $\rightarrow$ hub. A community with a cut road and no airstrip cannot be reached. With these costs, one job sent alone to Kintore is 92% travel, inside Nous's "up to 96%"; ten jobs on the trip bring it to 57% (Appendix D).

Settings differ in how each job is valued, in λ , and in whether shared trips are allowed:

1. **Cheapest jobs first:** every job is worth the same; $\lambda = 1$.
2. **Cheapest first + urgent guarantee:** as 1, plus a large bonus for an urgent job about to pass the equal clock (2 business days everywhere).
3. **Need first, official allowance:** need points, deadline bonus against FS17's remote clocks; $\lambda = 0.2$.
4. **Need first + urgent guarantee:** need points and the guarantee against the equal clock; $\lambda = 0.2$ or 0.5 .
5. **ReachNT:** setting 4 with H3 shared trips. Setting 1 is also run with shared trips.
6. **Need first without a guarantee:** λ from 0 to 0.8, to trace the whole curve.

2.6 Simulation, explanation and the ledger

Requests arrive at 4 per house per year, 25% more in the wet season, with faults drawn from the APY job mix; 30% use wording the reader was not trained on. One year (52 weeks plus 8 to clear the backlog) runs under each setting, with 73 tradespeople sized at 1.3 times expected job hours. A second report of the same fault at the same house, while the first is open, joins that job and is fixed on the same visit; an Immediate fault is made safe the day it is reported, and fixing it is a second clock. The solver is single-threaded with a fixed seed, so every run gives the same numbers.

How to read our numbers. *Cost per repair* is everything spent in the year (labour, travel, charters, overnight stays) divided by the repairs done. *9 in 10 fixed within* is the number of days by which 9 out of 10 urgent repairs were done (the 90th percentile). *Fault-days* add up every day a household lived with an unfixed fault; a dangerous fault counts more than a small one.

Each week a job is not done, the planner logs the reason: road cut, no trip because of cost, crew fully booked, or the trip came but did higher-scoring jobs first. The tenant's answer is assembled from that log, the score parts and the signed decision record, so it can only state reasons that happened. A test checks each section reads at Flesch 50 or above.

2.7 Personal information and the database

ReachNT will hold names, phone numbers and addresses, so the design keeps them in one place (Appendix F). A `pii` schema holds them encrypted with pgcrypto, readable only by intake staff through a function that logs every read. The `ops` schema holds work keyed by house ID and H3 cell. Row-level security limits a tradesperson to their trips and nearby open jobs, a tenant to their own jobs, and the public to resolution-6 summaries. The decision ledger is append-only, each row hashing the one before it. Everything runs on PostgreSQL, PostGIS and h3-pg, all free and open source: on NT Government servers for \$0 in licences, or for a pilot on Neon's Sydney region, which supports all three; its free tier has 1 GB and 100 compute-hours a project [22]. A year of data is under 1 GB.

2.8 Inclusive intake and decisions

Household points (a baby, an elder, a crowded house, a repeat) used to come only from the tenant's words, so a household could lose up to 45 points by saying less. Now whoever the tenant tells asks the same six questions, with a free interpreter where needed (Aboriginal Interpreter Service, TIS National, National Relay Service); the tenancy record gives household size and the job history gives repeats. Any "yes" counts; an unanswered question never removes points. The clock starts when the tenant first told anyone. Channel, language and timing are never scored (a unit test checks). A person can raise urgency at any time; lowering a dangerous repair needs someone who spoke to the tenant or saw it. A missed visit goes to the next trip, a named crew or any contractor of that trade, keeping its wait and deadline boost (Appendix I).

3 Findings

3.1 Left alone, efficiency pushes remote repairs to the back

Under cheapest-jobs-first, town tenants got 90% of urgent repairs within 3 days. Remote tenants waited up to 62 days, and 61% of their urgent repairs passed the 2-day equal clock (45% passed even the official 5-day remote clock). Fault-days, the danger of each fault times the days it stayed unfixed, totalled 226,000, and 821 jobs were still open after the extra eight weeks. No setting cost less per job, which is why it looks reasonable on a cost report.

3.2 The trade-off, priced

Figure 2 places every setting on two charts. ReachNT cost \$803 per repair against \$559, an extra \$244 per repair or \$9.6 million a year across the five hubs. For that, fault-days fell 73% (226,000 to 61,000) and the time to fix 9 in 10 urgent remote repairs fell from 62 days to 3. The guarantee protects urgent jobs: need ranking without it ($\lambda = 0.4$) cost \$703 per repair and left urgent remote repairs at 17 days. Ignoring cost altogether cost \$1,025 per repair for 5% fewer fault-days than ReachNT.

3.3 What shared trips changed

Against the same setting without them, shared trips cut cost per repair from \$876 to \$803 and fault-days from 74,300 to 61,000. Communities cut in the wet went from 7 to 3 days at the 90th percentile for urgent repairs. With a tighter budget ($\lambda = 0.5$), ReachNT cost \$751 per repair, less than cheapest-first with a guarantee (\$759), with fewer fault-days (76,000 against 88,000). Even cheapest-first gains from shared trips (226,000 to 179,000 fault-days).

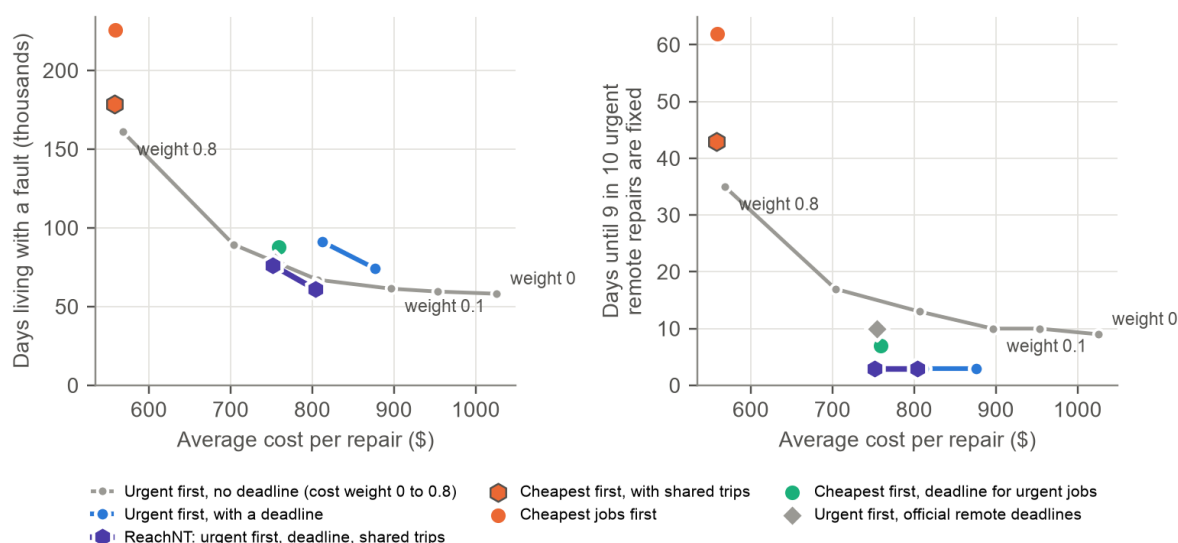


Figure 2. Average cost per repair against total fault-days (left) and against the remote urgent wait (right): days until 9 in 10 urgent remote repairs are fixed. One synthetic year, whole NT. Hexagon markers use shared trips.

3.4 Where the wait lands

Under cheapest-first, communities cut in the wet waited up to 69 days for 9 in 10 urgent repairs, and islands 73 (Figure 3). ReachNT brought these to 3 and 3. The hardest to serve are small, very remote road communities, where 90% of urgent repairs still took up to 14 days: few jobs share the cost of a long drive, and few have a neighbour within two rings.

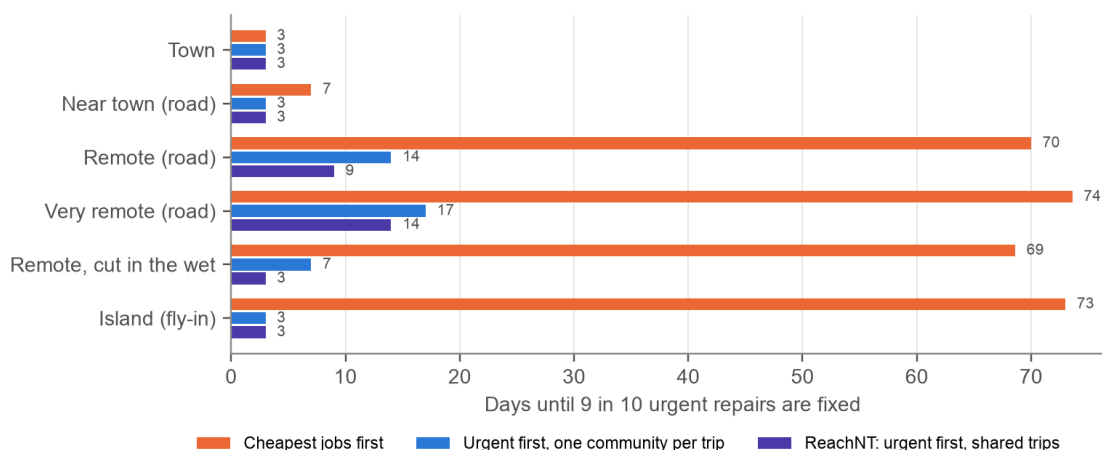


Figure 3. Days until 9 in 10 urgent repairs are fixed, by how hard the place is to reach.

3.5 Why jobs waited

The reason log (Figure 4) separates causes that look the same to a tenant. Under cheapest-first, most remote waiting was a cost decision. Under ReachNT, most of it was a fully booked crew, which calls for more trades or local trades, and a different answer to the tenant.

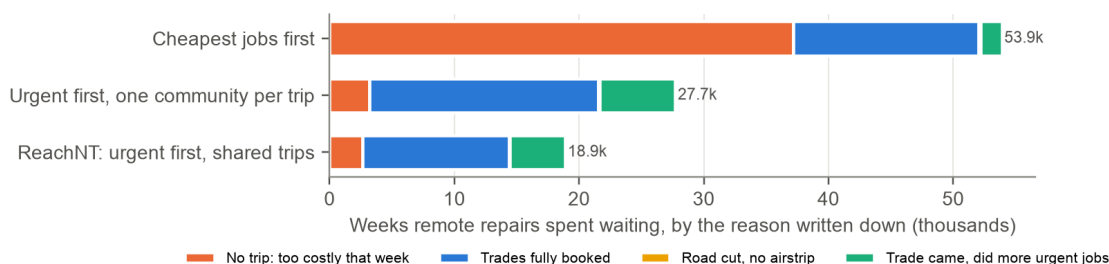


Figure 4. Job-weeks remote jobs spent waiting, by the reason the planner logged.

3.6 Reading free text is the weak link

Table 2. Reader accuracy on synthetic reports. "New wording" uses phrasings kept out of training.

Measure	Familiar wording (345)	New wording (690)
Spotting a dangerous fault, model ROC-AUC (0.5 = guessing)	1.00	0.84
Spotting a dangerous fault, model PR-AUC (guessing = 0.22)	0.99	0.72
Category correct, rules only	81%	61%
Category correct, model only	94%	63%
Category correct, combined	89%	80%
Under-triaged and not sent to a person	0.6%	3.3%
Dangerous fault missed and not sent to a person	0.0%	0.7%
Sent to a person	17%	40%

On new wording the model alone is only moderately good at spotting danger, and its confidence is not honest. The safety net works only because it sends four in ten reports to a person. Appendix G has the full measures: calibration, cross-validation, ranking agreement, solver gaps, and the results in four more random years. Real reports will come in Aboriginal English, Kriol and other languages [20], which synthetic English cannot test.

3.7 Sensitivity

We re-ran cheapest-first, need first with a guarantee, and ReachNT with crew capacity at 1.15 and 1.5 times job hours, the wet-season cut share at 0.2 and 0.5, the charter rate at \$1,500 and \$3,000 an hour, and a second random year (7 variants each). Cheapest-first ranged from \$536 to \$564 a job, with the remote urgent 90th percentile at 56–74 days; ReachNT ranged from \$741 to \$901 a job and 3 days in every variant. In every variant ReachNT had fewer harm-days and shorter remote urgent waits than cheapest-first. Run zones lowered both cost and harm-days in 7 of 7 variants. The direction of the trade-off holds; its size moves with the assumptions.

3.8 Saying less, and missed visits

We described 3,000 synthetic households in full and in a few words. From the words alone, the short telling lost 19 points on average, ranked lower 97% of the time, and only 4% of vulnerable households were recognised. With the standard questions it lost 1.8 and ranked lower 11% of the time (unanswered questions, assumed 1 in 10); 92% were recognised. When 1 in 10 booked visits missed, ReachNT still fixed 9 in 10 of the missed urgent remote repairs within 20 days, against 126 under cheapest-first, because a missed job keeps its clock and gets the deadline boost. Misses cost ReachNT \$58 per repair and moved its remote urgent 90th percentile from 3 to 9 days.

3.9 Fixes that didn't hold, trades together, and floods

A repair the tenant says is still broken reopens with 50 rework points and keeps its first day, so the deadline boost puts it on the next trip. Different trades booked to the same community in the same week can share one ute or charter: counted after planning, that saved ReachNT a further \$130 per repair. In a modelled flood of Kalkarindji, Daguragu and Pigeon Hole (94 repairs, roads cut for four weeks), the usual crews fixed 9 in 10 urgent flood repairs within 15 days, and the rest of the hub slipped from 3 to 10; with surge crews from the panel both stayed at 3.

4 Discussion

4.1 Ethical impacts

The brief names the risk: efficiency quietly decides who waits. ReachNT answers it three ways. The urgency score cannot see place. The equity cost is shown in dollars and days before anyone chooses. The choice is signed, so a tenant told "this was a cost decision" can see who made it and when. This follows the Robodebt Royal Commission's call for plain-language information about automated decisions and a path to review, and the Ombudsman's better practice guide [17]. ReachNT recommends and a person decides. Immediate jobs are always confirmed by phone and made safe by a local officer.

The score uses what is known about the household (a baby, an elder, a crowded house). Taken from the tenant's words alone, that rewarded people who say more; ReachNT takes it from the same questions on every channel, the tenancy record and the job history (§2.8, §3.8), and no one gains a place by reporting earlier, online or in better English. Privacy Act transparency rules for automated decisions start on 10 December 2026 for organisations they cover, which may include contractors [18].

4.2 Cultural and community impacts

Most remote tenants are Aboriginal, and many speak Kriol, Aboriginal English or another Aboriginal language first [20]. A reader trained on standard English will under-read them, as Table 2 shows on unfamiliar wording. ReachNT is an assistant to intake staff, never machine-translates into Aboriginal languages, and writes answers to be read out with a Community Housing Officer or interpreter.

Community-level results are sensitive: a map of long waits can read as a judgement on the community rather than the service. Under the CARE principles and Closing the Gap Priority

Reform 4, communities should see the data the Department uses, first [19]. The portal uses satellite imagery of Country rather than Aboriginal art; any artwork would be commissioned and licensed through an Aboriginal art centre under the Creative Australia protocols [23]. No community has seen ReachNT. Co-design with tenants, Aboriginal Housing NT, land councils and the Aboriginal Business Enterprises that hold maintenance contracts [9] must come before any use.

4.3 Limitations

The requests are synthetic, so absolute numbers are illustrations and the comparisons between settings are the finding. Travel uses straight-line distance at a blended road speed, and wet-season cuts outside the register are an assumption. Weekly planning means even town jobs can miss a 2-business-day clock by a few days. Shared trips pair two communities; longer chains would save more. A pilot needs work-order history from the Department's TMS and ASNEX systems [9].

5 Recommendations

For DHLGCD:

1. Keep urgency and logistics as separate, published steps; ask every tenant the same short questions on every channel, start the clock at first contact, and test that location, channel and language are not inputs to urgency.
2. Make the remote allowance in FS17 an explicit, signed decision, reviewed each quarter with its cost and its effect on remote waits, and report urgent waits by access band.
3. Plan trips on H3 shared trips, and give Community Housing Officers and tenants read access to each job's status and logged reasons, which closes the gap the ANAO found [10].
4. Build on PostgreSQL, PostGIS and h3-pg, hosted in Australia, with personal details in an encrypted vault and public outputs at resolution 6.

For contractors and industry:

5. Log a reason for every week a job waits and evidence for every missed visit, so cost decisions can be told apart from capacity shortfalls and a missed job is reassigned without losing its place.
6. Price charters, shared trips, trades travelling together and a disaster surge panel into contracts, and train local tradespeople in the communities that wait longest.

For communities and their organisations:

7. Co-design the tenant answer, the intake questions and the fault vocabulary in local languages with tenants, Aboriginal Housing NT and the Aboriginal Interpreter Service, under ethics approval and land council research permits.
8. Hold the decision ledger to account through local authorities and tenant groups, and give every tenant a way to ask a person to review their ranking, answered within 10 working days.

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- [22] Neon. Postgres extensions; regions; pricing. <https://neon.com/docs/extensions/pg-extensions> ; <https://neon.com/docs/introduction/regions> ; <https://neon.com/pricing> ; h3-pg: <https://github.com/zachasme/h3-pg>
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Appendix A: AI usage declaration

We used Claude (Anthropic) as a coding and writing assistant: to search for and summarise public sources, draft and review Python and JavaScript, and draft and edit report text. The NT sources behind the figures we quote (DHLGCD FS17, the NIAA review, Nous 2017, Grealy et al. 2022, the Menzies Healthy Homes evaluation, the ANAO audit) were read in full and the figures checked against them (links in research/sources/README.md). Other references, mainly overseas methods and legal context, were checked through their published summaries. All numbers in this report come from run_all.py through outputs/numbers.json. The team reviewed, tested and is responsible for all content. ReachNT itself uses no large language model: the reader is phrase rules plus a logistic regression, and tenant answers are assembled from logged facts.

Appendix B: Code, portal and how to run

```
pip install -r requirements.txt
python run_all.py rebuilds every number, figure and the portal data (about 15 minutes on 8 cores)
python run_all.py --quality quality measures only (Appendix G)
python run_all.py --extras the inclusion measure and the missed-visit experiment (section 3.8)
pytest -q tests 75 checks: need-only urgency, inclusive intake, reader safety net, planner limits, missed visits, data integrity
node tests/test_api.mjs 17 checks on the server: validation, no-access evidence, urgency changes, reassignment, sign-in rules
https://reachnt.vercel.app the ReachNT portal, desktop and phone (or serve web/ locally); works offline
https://github.com/harshrastogii/reachnt all code, data links and documents
notebooks/01_walkthrough.ipynb the method step by step
```

Code: src/reachnt/ geo.py (communities, access, H3 cells and shared trips), synth.py, intake.py (reader), urgency.py (need score), planner.py (CP-SAT trips and shared trips), simulate.py, explain.py (tenant answers), experiments.py, evaluate.py (quality measures), figures.py, export_web.py. Parameters, each tagged as sourced or assumed: config/params.yaml. Fault taxonomy and the standard intake questions: config/taxonomy.yaml. Database: docs/schema.sql and docs/DATABASE.md. Inclusive decision model: docs/INCLUSIVE_DECISION_MODEL.md.

Appendix C: Data sources and dataset links

Table C1. Data sources. Job-level requests are synthetic; everything else is public.

Source	What we take from it
NIAA/PwC's Indigenous Consulting (2023), Review of the NPRH NT, App. B [2]	70 communities; houses estimated as overcrowded houses ÷ overcrowded share (4,481)
NTG remote communities list 2021 (CC BY)	Coordinates and land council for each community
ABS 2021 Census, Indigenous Profile I13 (CC BY 4.0)	Households renting from the housing authority in the five hub towns (3,864)
NT Annual Traffic Report 2023 restriction register (CC BY)	Wet-season closures for 14 communities, e.g. Lajamanu Road closed Nov–Apr
Road Report NT live feed; OurAirports	Current closures; airstrips within 8 km
DHLGCD FS17 [5]; RTA 1999 (NT) s 63 [16]; Healthabitat [8]	Response clocks; emergency repairs in law; order of health impact
Grealy et al. 2022, Table 3 [7]; Nous 2017 [6]	Job mix by trade (38,492 APY jobs); cost structure for calibration
Digital Earth Australia Landsat geomedian 2024; Esri World Imagery	Satellite basemap in the portal (offline and online)

NTG remote communities list:

<https://data.nt.gov.au/dataset/remote-communities-with-mobile-coverage>

ABS 2021 Census DataPacks (Indigenous Profile, ILOC):

<https://www.abs.gov.au/census/find-census-data/datapacks>

NT Annual Traffic Report 2023: <https://data.nt.gov.au/dataset/annual-traffic-report-2023>

Road Report NT: <https://roadreport.nt.gov.au>

OurAirports: <https://ourairports.com/data/>

Digital Earth Australia Landsat geomedian: <https://www.dea.ga.gov.au>

ABS ASGS 2021 boundaries (coastline): <https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3>

Appendix D: Assumptions and calibration

Table D1. Cost calibration. One 2.5-hour job sent alone, and ten jobs sharing one trip, against Nous (2017): travel up to 96% of an emergency job's cost, 11–37% for planned work.

Comm unity	Mode	One job: total	One job: travel share	Ten jobs: travel share	Ten jobs: cost per job
Kintore	road	\$3,911	92%	57%	\$748
Lajama nu	road	\$3,488	91%	54%	\$705
Wadeye	road	\$2,377	86%	45%	\$594
Galiwin ku	air	\$4,784	93%	61%	\$835
Barung a	road	\$793	59%	13%	\$372

Main assumptions (all in config/params.yaml): labour \$130 an hour; vehicle \$1.10 a km; overnight \$320; charter \$2,200 an hour; blended road speed 70 km/h with 1.2 circuitry; requests 4 per house per year; crews 1.3 × expected job hours; 35% of wet-season weeks cut for Top End communities more than 2 hours out that the register does not cover.

Appendix E: A tenant's answer (synthetic)

An urgent job at the 90th percentile of waits in a community cut in the wet, under cheapest-first (J37458; community name withheld). Text message: "Your no water to the house repair is urgent. Waiting 62 days. Held up by travel cost. Ask why: 1800 104 076"

What we heard. You told us: "solar hot water not working - when can you come - grandkids staying with us". We read this as: no water to the house.

How urgent it is. This is an Urgent repair. Our rule is the same in town and out bush: a plumber within 2 business days. The law calls this kind of fault an emergency repair.

Where it is up to. Not fixed yet. You have waited 62 days. That is longer than our rule, by 59 days.

Why it is not fixed yet. For 56 days no plumber was sent to [community]. It costs about \$1,800 to drive one there and back. That was a cost decision, not a judgement about your repair. Nobody signed off on it. For 7 days every plumber was fully booked on repairs that scored higher than yours. Higher scores mean more danger or a longer wait.

What would change it. If it gets worse (sparks, smoke, water everywhere, sewage), call 1800 104 076 or tell your housing officer. It becomes Immediate and is made safe the same day.

Every day you wait adds points, so your repair moves up the line.

Your rights. You can ask for a free interpreter at any time. If an urgent repair is not done, you can ask the Tribunal (NTCAT) for an emergency repair order. You can also complain to the NT Ombudsman. You can ask for a person to review how your repair was ranked: say so when you call, tell your housing officer, or use the app. We answer within 10 working days.

Score 813: 500 for the urgent category, 70 for harm, 36 for the health practice it affects, 25 for who lives there, 0 for a repeat report, 182 for waiting. Distance and cost are never part of this score.

Appendix F: Who sees what

Table F1. Data zones and access in the ReachNT database (docs/schema.sql).

Zone	Holds	Who can read it
pii (vault)	Name, phone, address, language, interpreter need; all personal fields encrypted	Intake staff through a logged function; the tenant through a one-time code
ops (work)	Houses as ID + H3 res-10 cell, jobs, scores, trips, weekly reasons, decision ledger	Coordinator (own hub); tradesperson (own trips + open jobs within 3 res-5 rings); housing officer (own communities); analyst (no vault)
public (views)	Waits and fault-days by H3 res-6 cell and access band, cells under 5 jobs hidden	Anyone

Appendix G: How good are the algorithms?

Every measure below is computed by src/reachnt/evaluate.py on synthetic reports: 345 in familiar wording and 690 in wording kept out of training, 22% of them dangerous (Immediate). ROC-AUC is the chance the model scores a random dangerous report above a random safe one (0.5 is guessing, 1 is perfect). PR-AUC summarises how many flagged reports are truly dangerous across all thresholds; guessing scores the share of dangerous reports, 0.22. Ranges are 95% bootstrap intervals.

Table G1. Reader quality. "Model" is the logistic regression alone; "system" is rules, model and the send-to-a-person checks together.

Measure	Familiar wording	New wording
Dangerous fault, ROC-AUC (model)	1.00 (0.99–1.00)	0.84 (0.80–0.88)
Dangerous fault, PR-AUC (model)	0.99 (0.98–1.00)	0.72 (0.65–0.78)
Dangerous reports caught: labelled Immediate or sent to a person (system recall)	100%	99%
Dangerous reports labelled Immediate straight away	96%	87%
Flagged reports that were truly dangerous (system precision)	54%	39%
Category (Immediate/Urgent/Routine), macro ROC-AUC (model)	1.00	0.80
Category, macro PR-AUC (model)	0.99	0.69
Category, accuracy / macro-F1 (system)	89% / 0.89	80% / 0.80
Category, quadratic-weighted kappa (system)	0.83	0.72
Exact fault type (23 types), top-1 / top-3 accuracy (model)	92% / 100%	47% / 72%
Exact fault type, macro-F1 (model)	0.92	0.46
Calibration: Brier score / expected calibration error (model, lower is better)	0.13 / 0.09	0.76 / 0.16

Five-fold cross-validation on the training reports gave macro-F1 0.93 ± 0.02 and danger ROC-AUC 1.00 ± 0.00 , so the model is stable on wording it knows. The drop on new wording is the real finding: the model alone separates dangerous from safe reports only moderately (ROC-AUC 0.84), and its confidence runs ahead of its accuracy at the top (Figure 5, right). The system still catches 99% of dangerous reports because it sends doubtful ones to a person, at the price of a precision of 39%. For that reason ReachNT never shows tenants a confidence percentage.

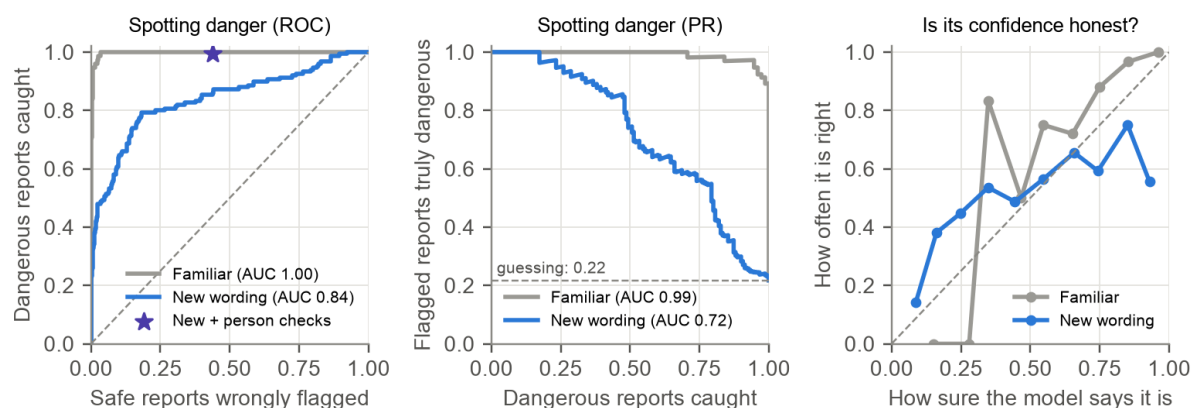


Figure 5. Left and centre: how well the model's danger score separates dangerous from safe reports; the star is the whole system on new wording, with a person checking doubtful reports. Right: when the model says it is X% sure, how often it is right (the dashed line is honest confidence).

Moving the confidence threshold from 0.30 to 0.90 changed the share sent to a person only between 39% and 41%: most referrals come from the danger checks, not the threshold. We kept 0.55, set before testing; tuning it on the test reports would flatter the result.

Ranking. On new wording, the queue built from read reports agreed with the queue built from the true faults at Kendall's $\tau = 0.62$ (0.85 once a person has corrected the reports sent to them). Of

the 150 reports at the top of the queue, 79% were truly dangerous before that check and 95% after.

Planner. Across the 25,996 weekly plans in the runs below, CP-SAT proved 99.4% of them optimal: no better plan exists under the model's rules. The other 0.6% stopped at the 2-second limit with a feasible plan; for those the solver can only bound how much better a plan might be, at most 85% in the worst week, and such bounds are usually loose. The median plan took 7 milliseconds and 95% took under 0.16 seconds.

Table G2. The headline plans in five random years of synthetic requests (the main year and four more). Mean, with the range.

Plan	Cost per repair	9 in 10 urgent remote repairs fixed within (days)	Fault-days (thousands)
Cheapest first	\$559 (\$558–\$560)	63 (62–65)	229 (226–231)
Urgent first, one community per trip	\$875 (\$867–\$882)	4 (3–6)	75 (70–79)
ReachNT	\$805 (\$795–\$816)	3 (3–3)	61 (58–63)

In every year ReachNT cut fault-days against cheapest-first by 73%–75%, for 42%–46% more per repair, and shared trips saved \$66–\$73 per repair against the same plan without them.

Appendix H: Validation and accountability for a pilot

We checked ReachNT against the standards similar services now have to meet. The full list, with sources, is in docs/VALIDATION.md. The live portal has a strict content security policy, serves its libraries from its own site, and checks every update it receives. It shows zero WCAG 2.2 AA issues in every view, on desktop and phone. All 70 community positions were checked against satellite imagery.

ReachNT now follows the rules that would apply to a pilot. A plain-language notice of its automated decisions, with an AI transparency statement, meets the Privacy Act's transparency duty from 10 December 2026 (web/privacy.html). An AI impact assessment follows the format of the Commonwealth AI policy v2.0 (docs/AI_IMPACT_ASSESSMENT.md). Tenants can ask a person to review their ranking, and a person re-reads a weekly sample of the reader's work, as the Robodebt Royal Commission and the Commonwealth Ombudsman's guide recommend. "Made safe" and "fixed" are separate clocks, as in England's Awaab's Law. No job closes on an unevidenced "no one home", a failure the UK Housing Ombudsman found in 2025.

A person can check and change urgency at any time, a missed job is reassigned without losing its clock, and household points come from the same questions on every channel (Appendix I). Before real use, the reader must be tested on real reports, by language, with the Aboriginal Interpreter Service, following docs/REAL_LANGUAGE_TEST_PROTOCOL.md.

Appendix I: The inclusive decision-making model

ReachNT's rule is that a repair's place in line follows need only: the fault, who lives in the house, and how long the household has waited. The model is written up in full in docs/INCLUSIVE_DECISION_MODEL.md.

Table I1. What used to disadvantage a tenant, and what ReachNT does instead.

Disadvantage	What ReachNT does
Saying less, limited English, a relayed message	The same six questions on every channel, read out by staff or an interpreter: anyone in danger now; a baby or young child; an elder; someone sick, pregnant or with a disability; how many people live there; told anyone before. The tenancy record gives household size (more than 2 people per bedroom is crowded) and the job history gives repeats. Any "yes" counts; "not asked" never removes points and triggers a call-back.
Language	Words kept as said. Free Aboriginal Interpreter Service, TIS National and National Relay Service. Language recorded only to book an interpreter, never scored.
Digital literacy, channel	Repairs line, Community Housing Officer, maintenance officer, tradesperson, front counter and app create the same record. Everything in the app can also be done by phone or through the housing officer.
Submission timing	The clock starts at first contact with anyone. The line is ordered by need; waiting adds 3 points a day only after half the clock, so no job waits forever and an early caller with a minor fault never jumps a dangerous one.
Unchecked machine decisions	A person can raise or confirm urgency at any time, from a call, a visit, a photo or a review. Lowering a dangerous repair needs someone who spoke to the tenant or saw the fault, and a reason the tenant reads. The reader can never lower danger. Enforced in the portal, the server and a database constraint.

When time runs short. The planner values a job by its need points, adds 400 when its clock ends within 7 days and, under ReachNT, 6,000 more for an urgent or dangerous job, which puts it on the next trip. The coordinator sees the time left and this arithmetic on every waiting repair.

Tenants without the app. The Community Housing Officer has their own view: they log reports with the same questions and record what a tenant tells them (fixed, still broken, it got worse, a review), only for their own communities, each marked as recorded for the tenant. "Still broken" reopens the job with rework points and its first day.

Floods, cyclones and fires. The coordinator declares an event over the communities hit. The portal plans the response (make every house safe within 48 hours, one team trip with the trades needed, surge crews from the contractor panel, a message to every household) and tags every job from the area, while need still sets the order and other communities keep their trips.

When a visit misses. The tradesperson records why ("No one home" and "Can't get in" need the time and what was tried). The job never closes. The coordinator sends it to the next trip, a named crew, or opens it to any contractor of that trade on the panel, including local Aboriginal Business Enterprises, where the first to accept gets it; "Needs another trade" changes the trade. A tenant or tradesperson can say it got worse, by phone, through the housing officer or in the app; a person calls back that day and records an urgency check. The tenant's timeline shows each check and each reassignment.